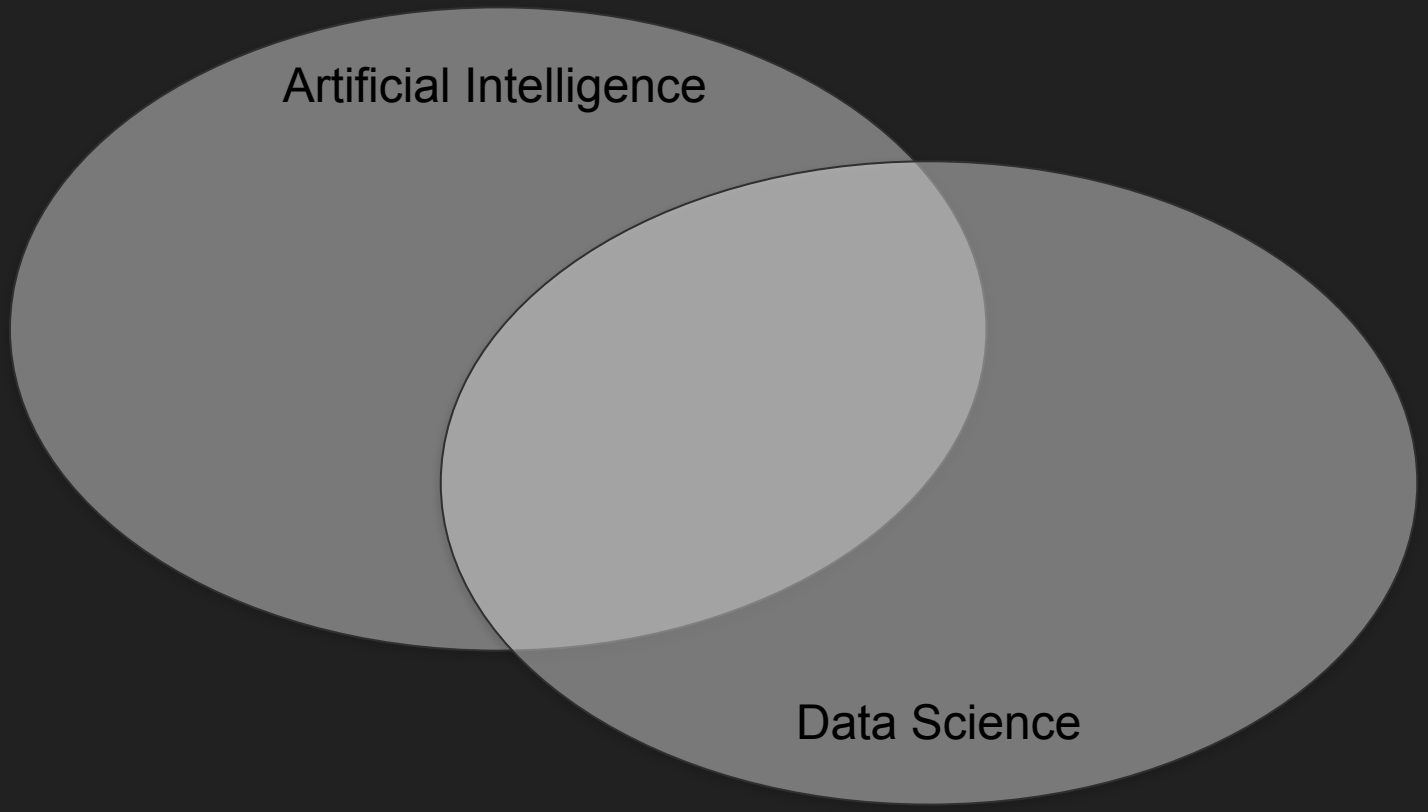
Hi!
I’m Teo!
A DPhil in Generative AI at University of Oxford

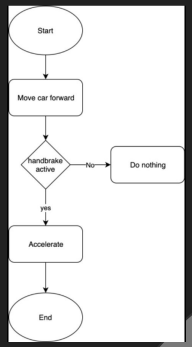
Reinforcement Learning Workshop

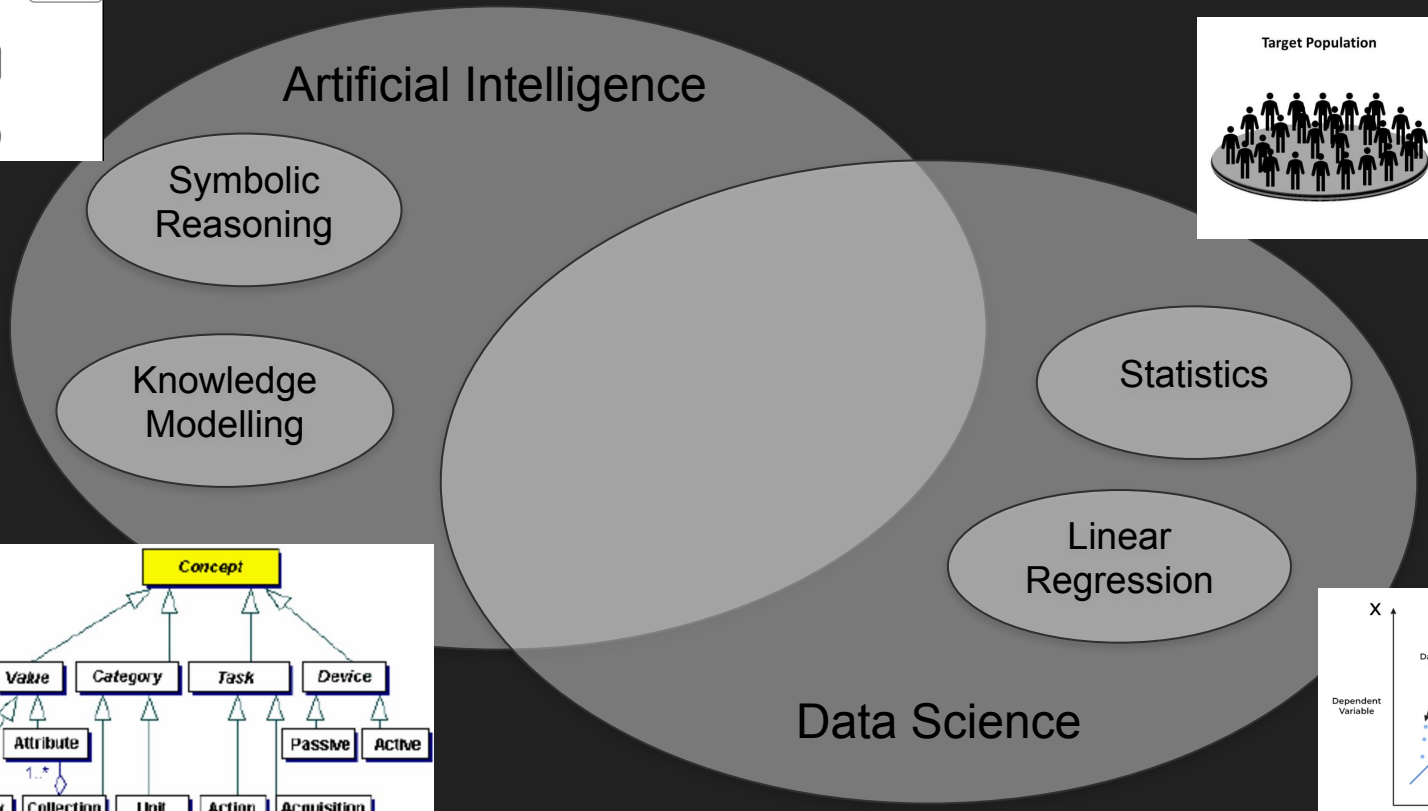
Fibonacci

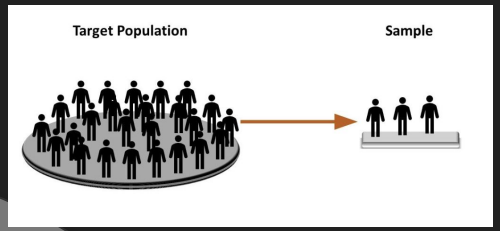
What will we do today

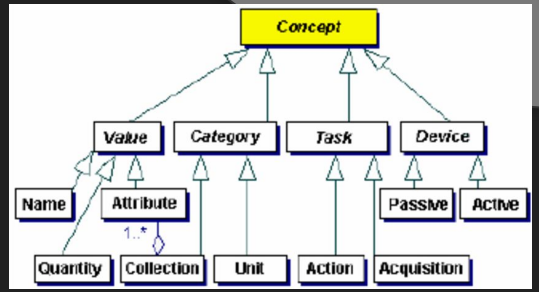
1. **Presentation** (40 minute)
2. **Break** (10 minute)
3. **Discussions and Questions** (10 minutes)
4. **Code presentation** (10 minutes)
5. **Walk through code & pair coding (50 minutes)**

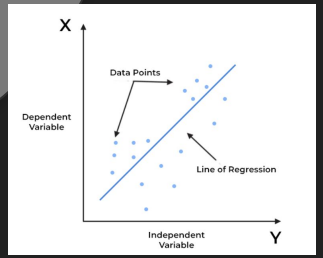

Artificial Intelligence
Data Science

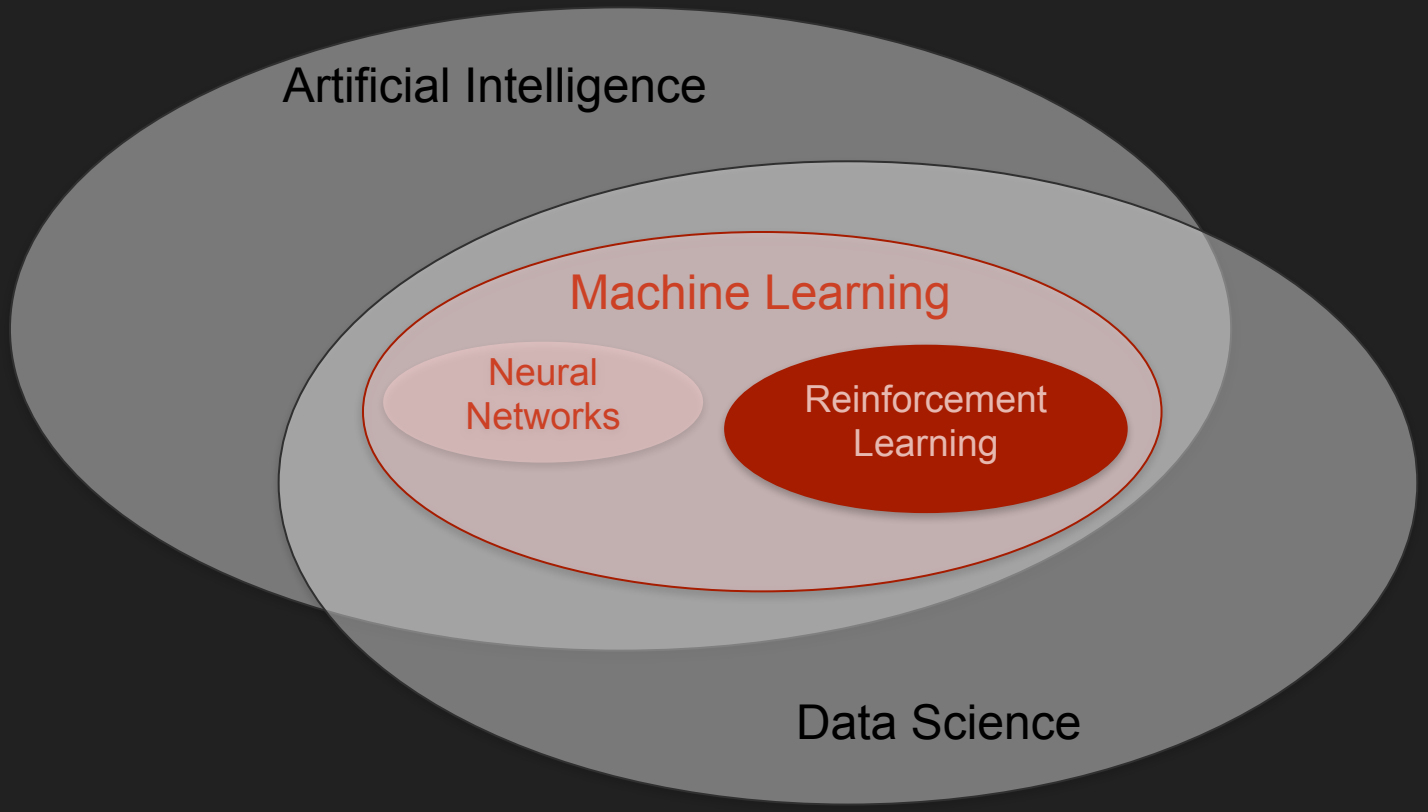

Start
Move car forward
handbrake active
No
Do nothing
yes
Accelerate
End


Artificial Intelligence
Symbolic Reasoning
Knowledge Modelling
Statistics
Linear Regression
Data Science


Target Population
Sample


Concept
Value
Category
Task
Device
Name
Attribute
1..*
Passive
Active
Quantity
Collection
Unit
Action
Acquisition

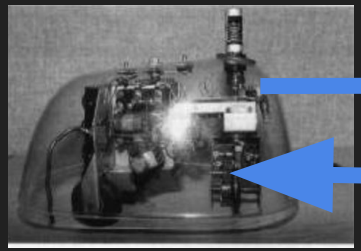

X
Data Points
Dependent Variable
Line of Regression
Independent Variable
Y

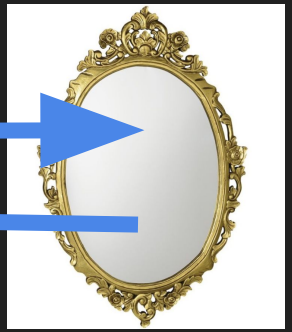

Artificial Intelligence
Machine Learning
Neural Networks
Reinforcement Learning
Data Science

Elmer and Elsie (1948)

- 2 electronic robots built from extra war materials and clockwork mechanisms, programmed to think.
- A light sensor.
- Programmed to avoid external obstacles using the sensors they were equipped with.

In one of the experiments, a light was placed on Elmer's shell, at which point the robot began to move erratically. The scientists saw this as a kind of enthusiasm.





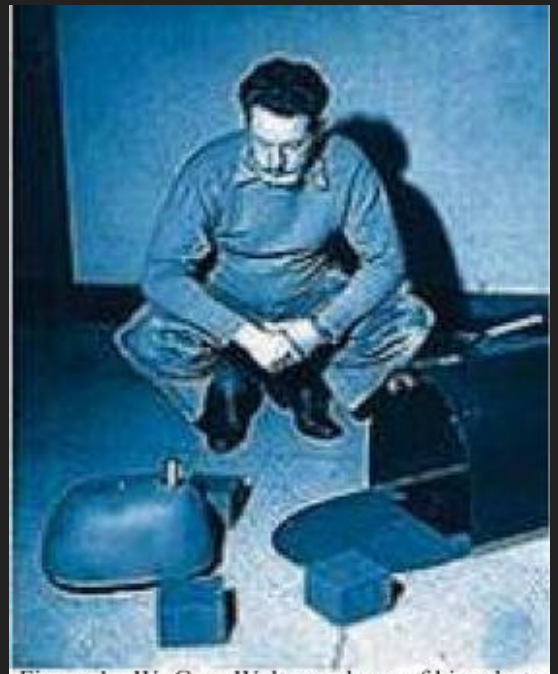
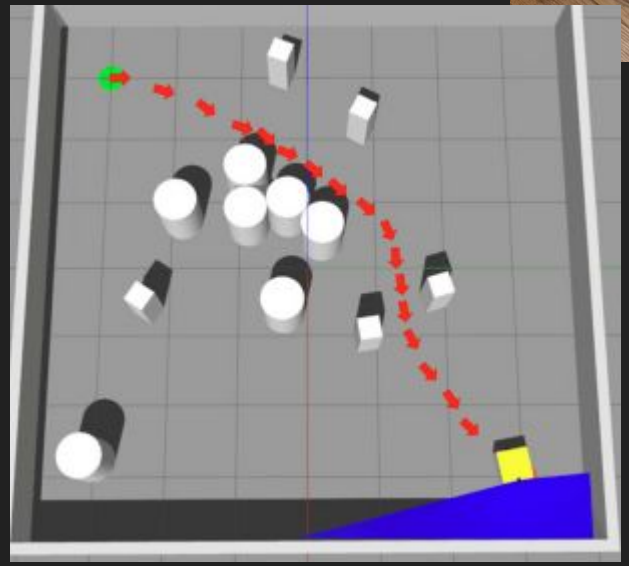


Figure 1 - W. Grey Walter and one of his robots

Hoover Robots (2010-)



Deep Blue vs. Kasparov (1996- 1997)

Deep Blue was a supercomputer built by IBM to play chess.

The 1996 match

Game #	White	Black	Result	Method of conclusion
1	**Deep Blue**	Kasparov	1–0	Resignation
2	**Kasparov**	Deep Blue	1–0	Resignation
3	Deep Blue	Kasparov	½–½	Draw by mutual agreement
4	Kasparov	Deep Blue	½–½	Draw by mutual agreement
5	Deep Blue	**Kasparov**	0–1	Resignation
6	**Kasparov**	Deep Blue	1–0	Resignation
Result: **Kasparov–Deep Blue: 4–2**				

The 1997 rematch

Game #	White	Black	Result	Method of conclusion
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Deep Blue
IBM chess computer

Garry Kasparov
World Chess Champion

Deep Blue vs. Kasparov (1996- 1997)

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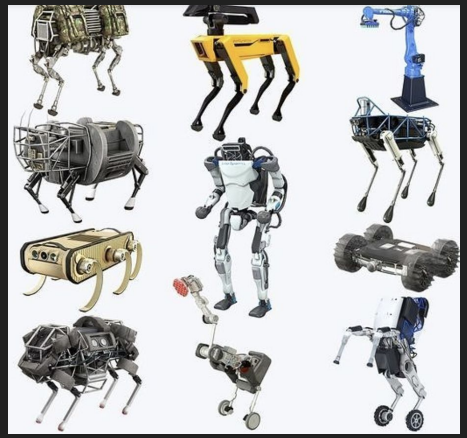
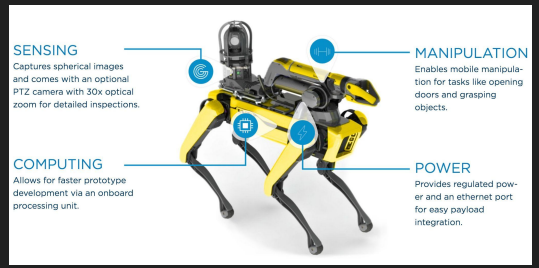
Deep Blue
IBM chess computer

Garry Kasparov
World Chess Champion

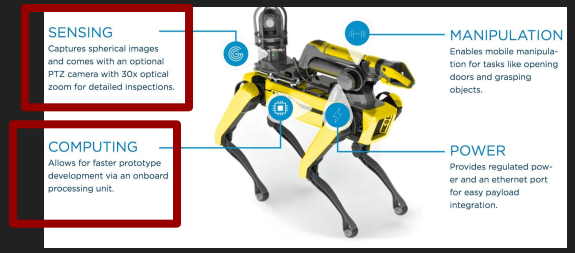
Boston Dynamics Robots (1992- 2024)

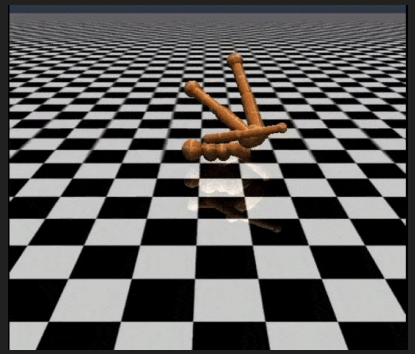
- Extremely agile, they navigate diverse terrains and adapt to obstacles.
- Used in industry for inspections, surveillance, and transport, demonstrating real-world applications of AI and reinforcement learning.
- They open doors and climb stairs, performing complex tasks and transforming interactions with the environment.



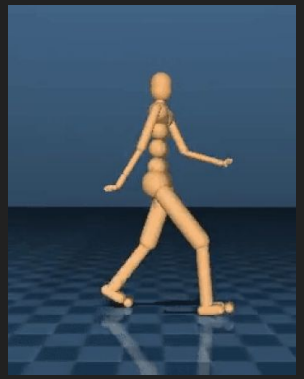


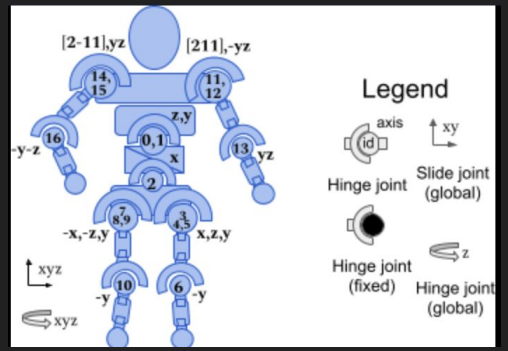
MuJoCo - DeepMind (2012)





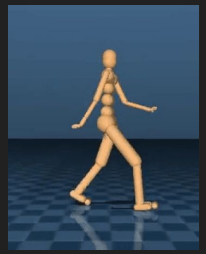
After Training



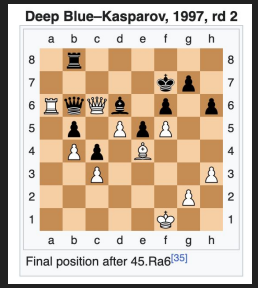


What did we observed?

Agent

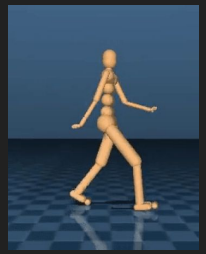


Environment

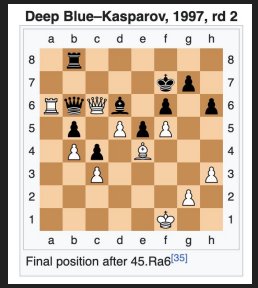


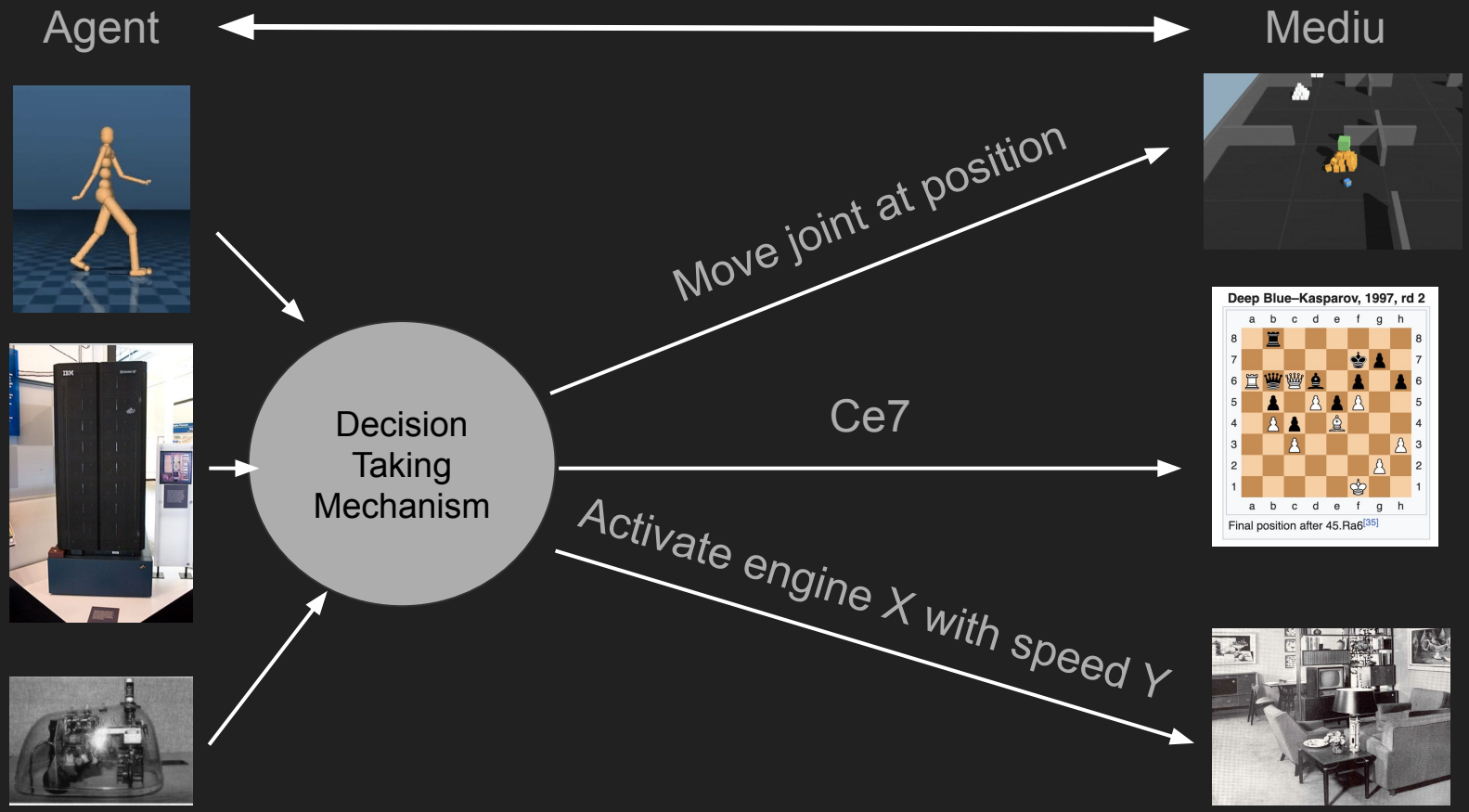
Continuous interaction

Agent



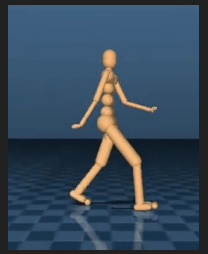
Mediu




Agent
Mediu
Decision
Taking
Mechanism
Move joint at position
Ce7
Activate engine X with speed Y
Deep Blue–Kasparov, 1997, rd 2
Final position after 45.Ra6[35]

Agent

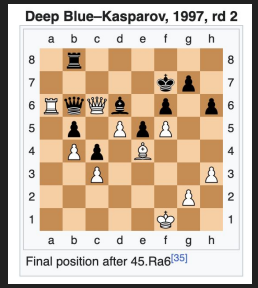
Mediu

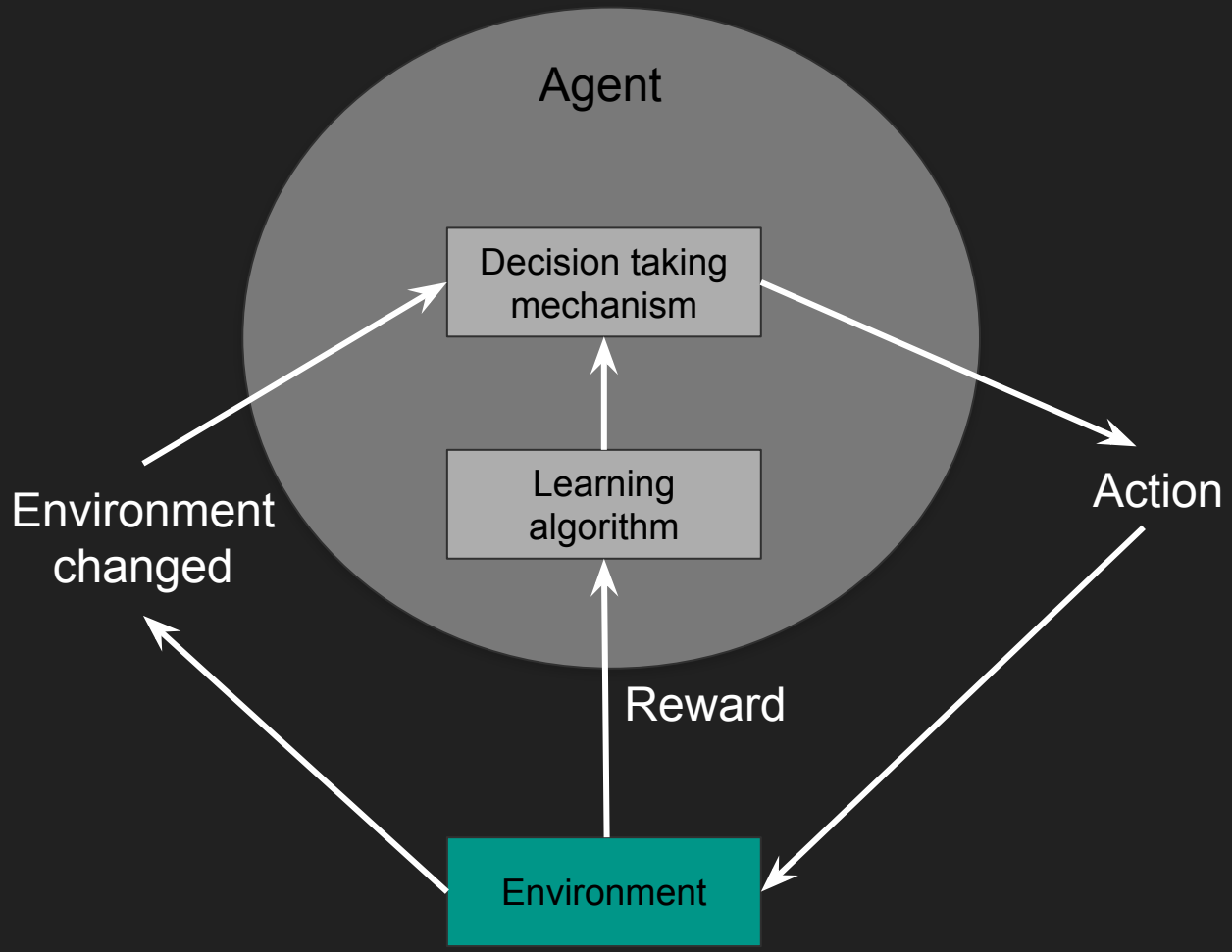


The agent touch the ground

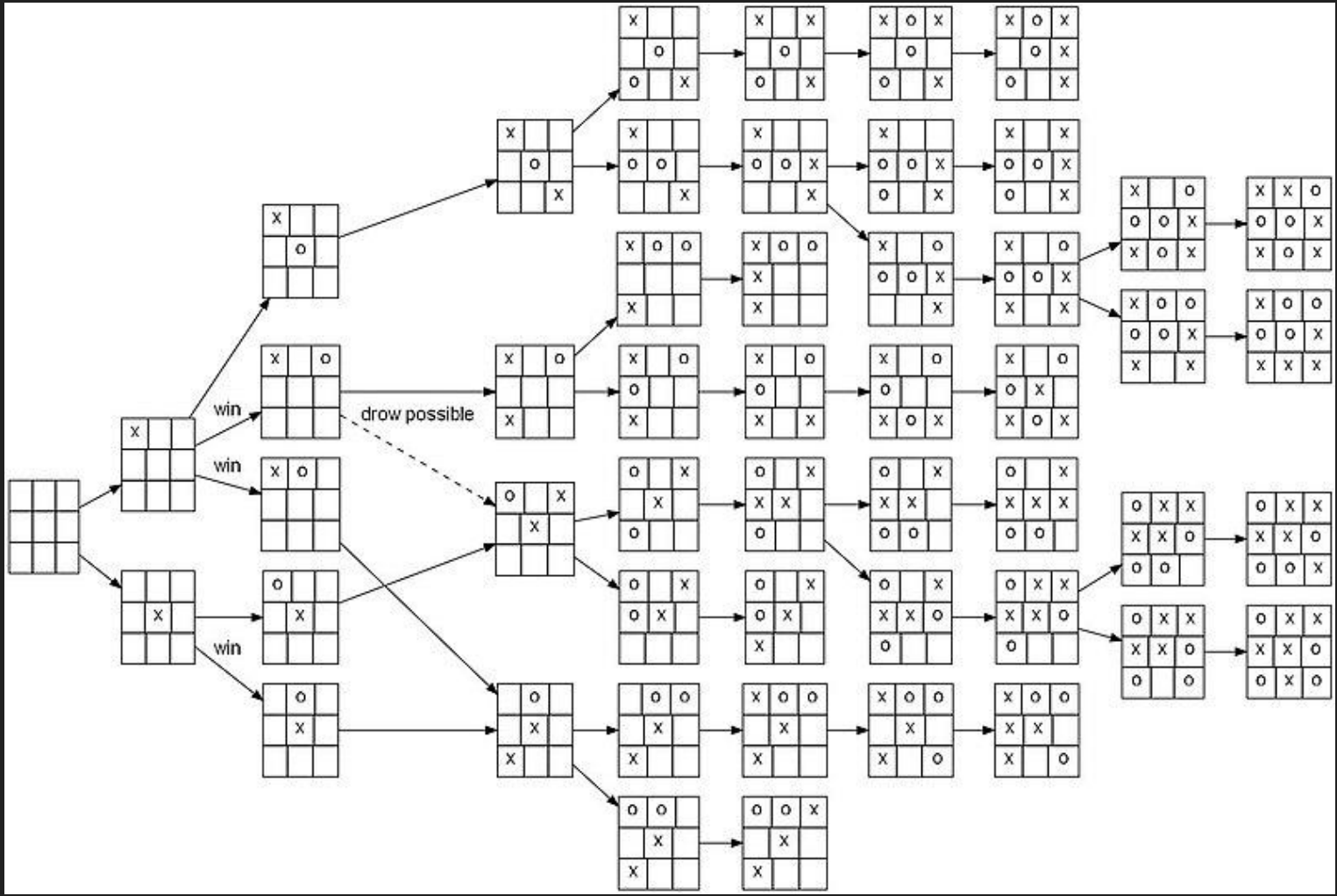
Train decision
Taking mechanism

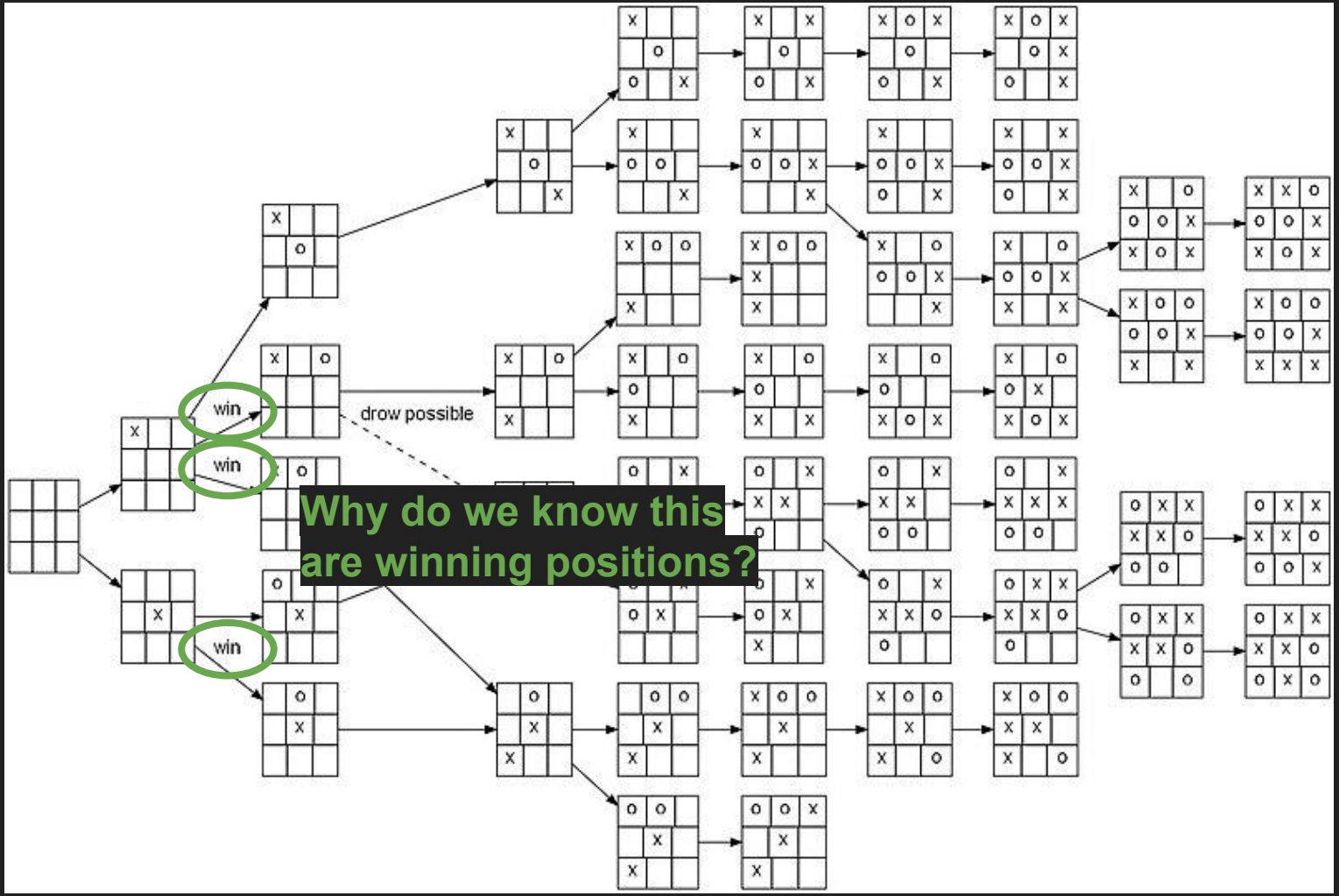
Decision
Taking
Mechanism



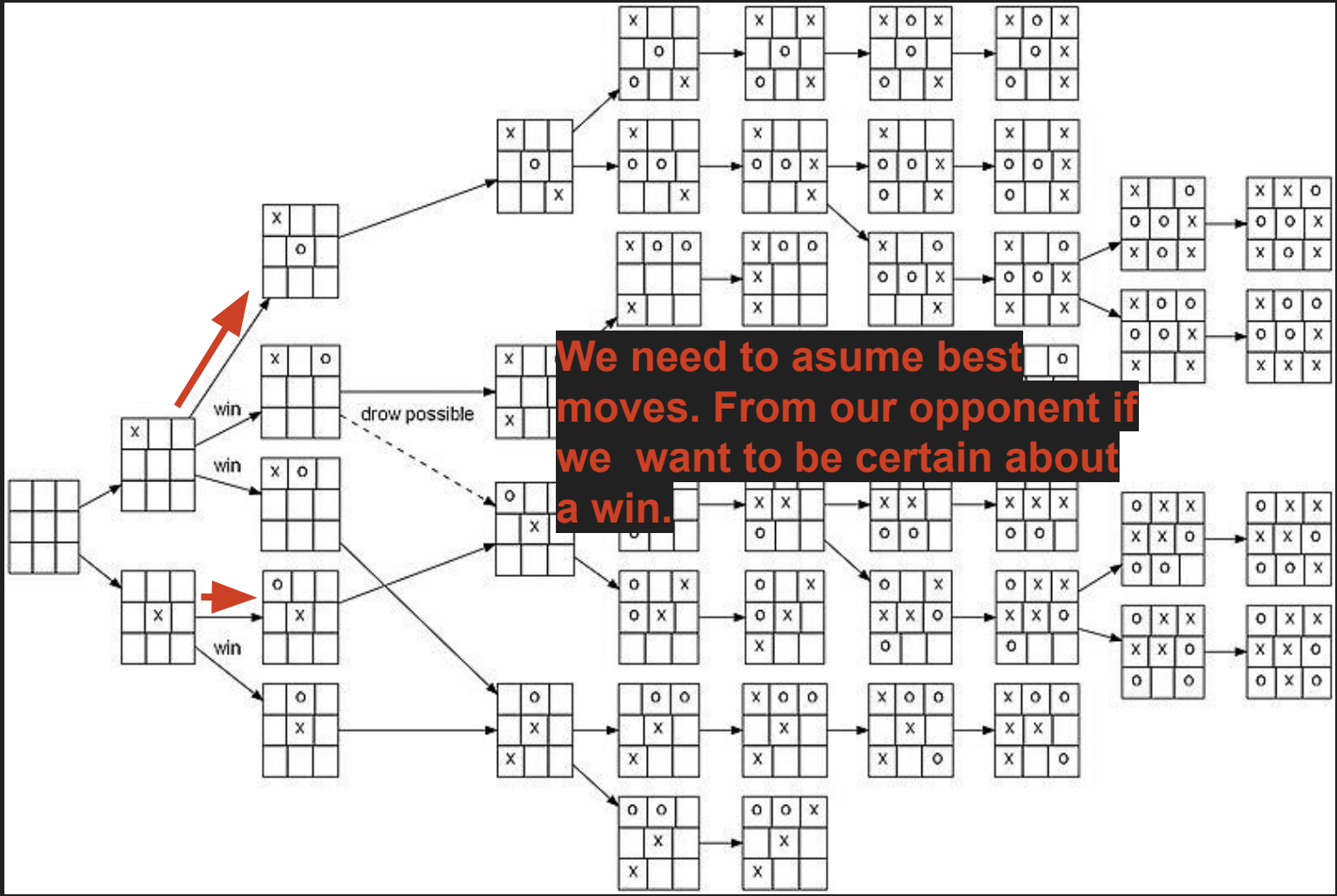

Agent
Decision taking mechanism
Learning algorithm
Environment changed
Action
Reward
Environment

How do we build “Intelligent” systems?


win
win
win
drow possible



Why do we know this are winning positions?

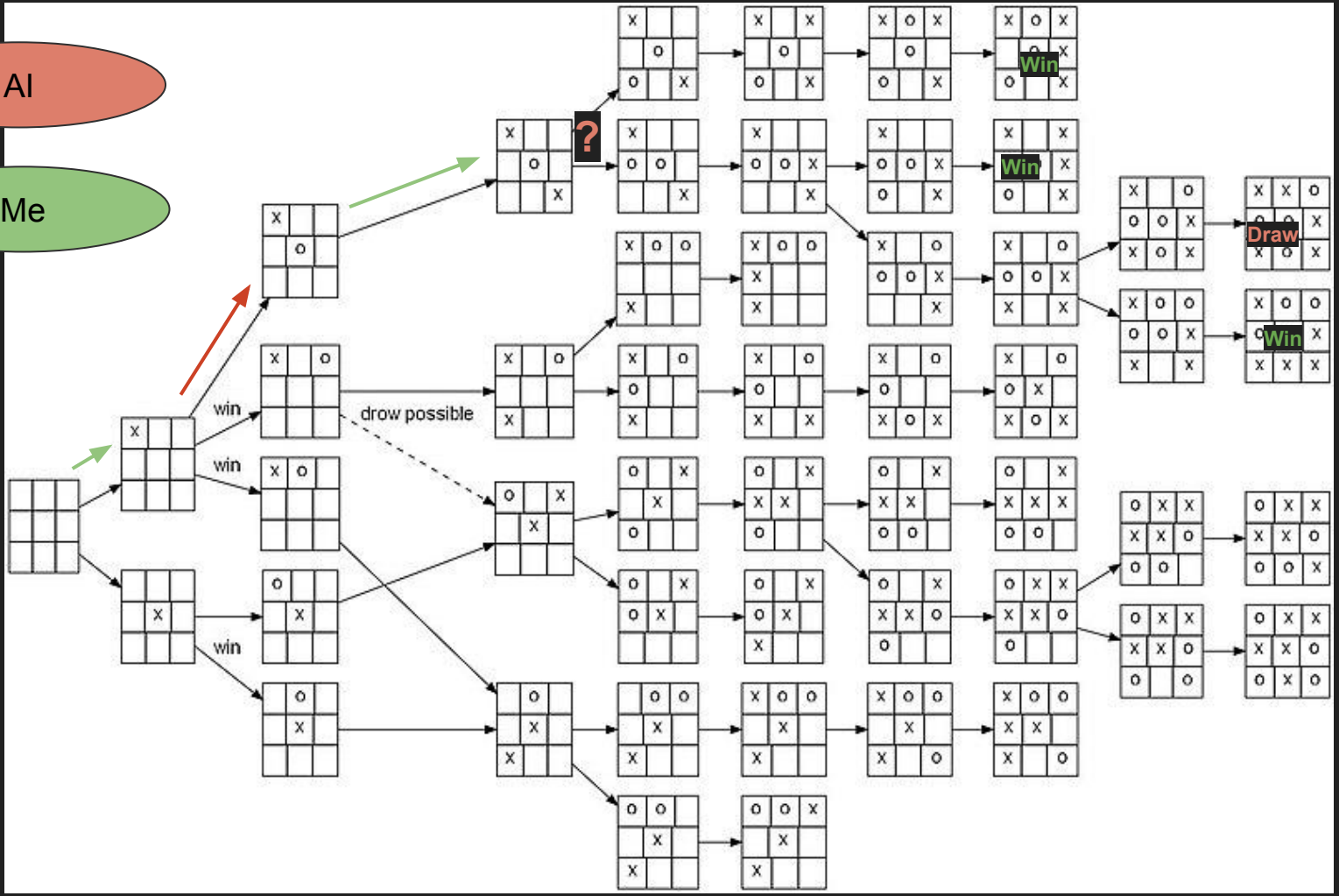


We need to asume best moves. From our opponent if we want to be certain about a win.

Let’s simulate a game!

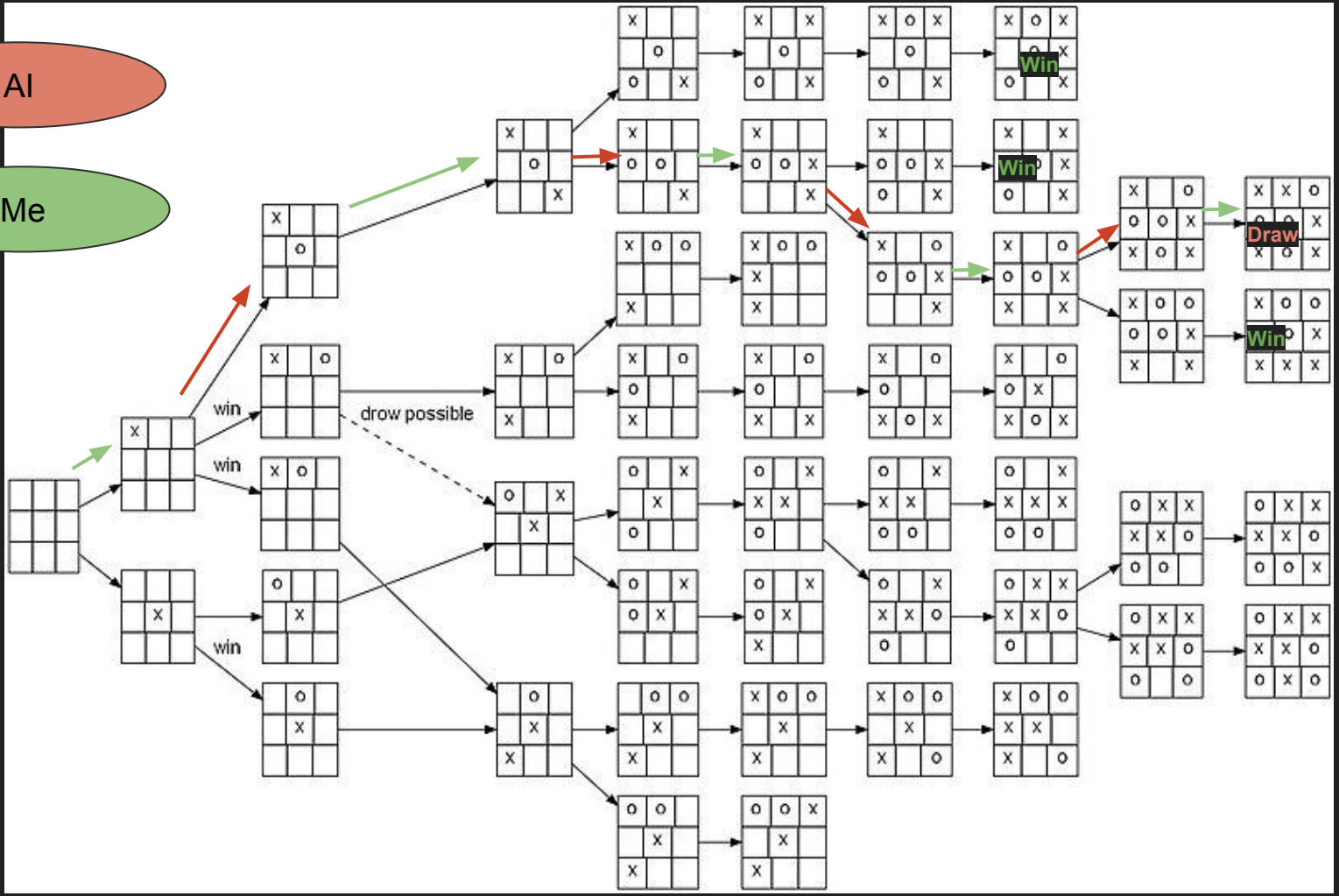
AI

Me



AI

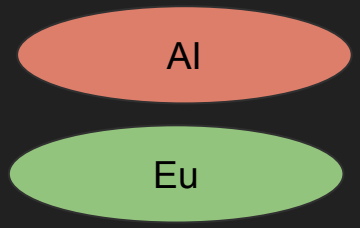
Me



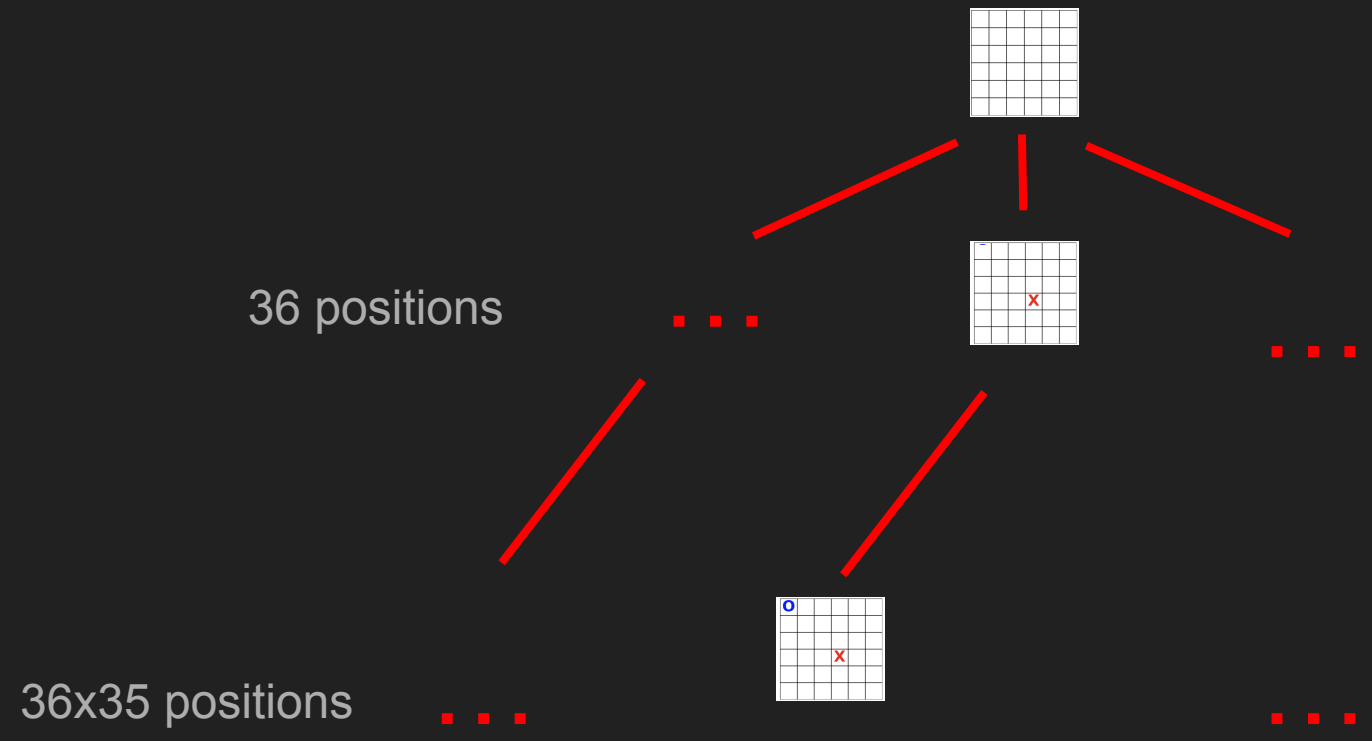
Observations:

1. What happens if we can't see until the end of the game?

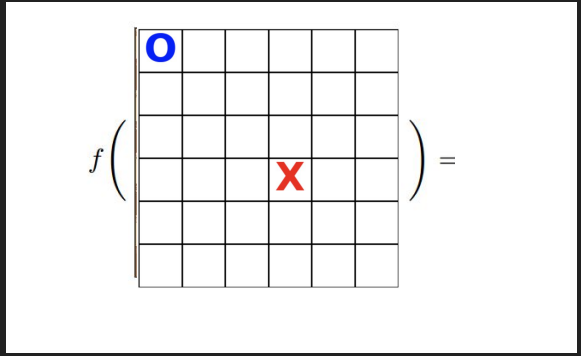
2. Mini-Max Algorithm

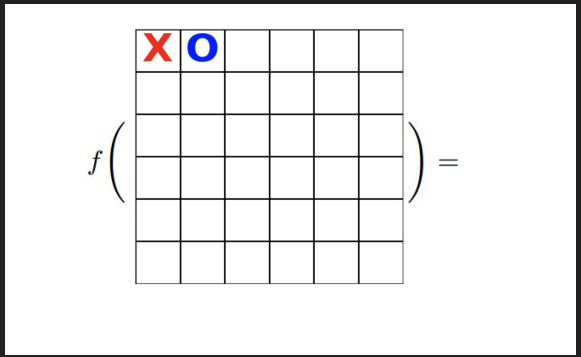


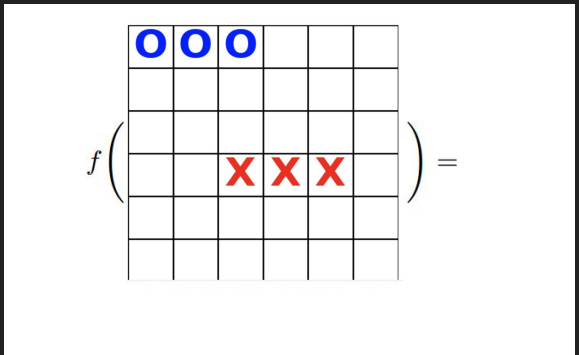
Number of positions on a 6 by 6 board.



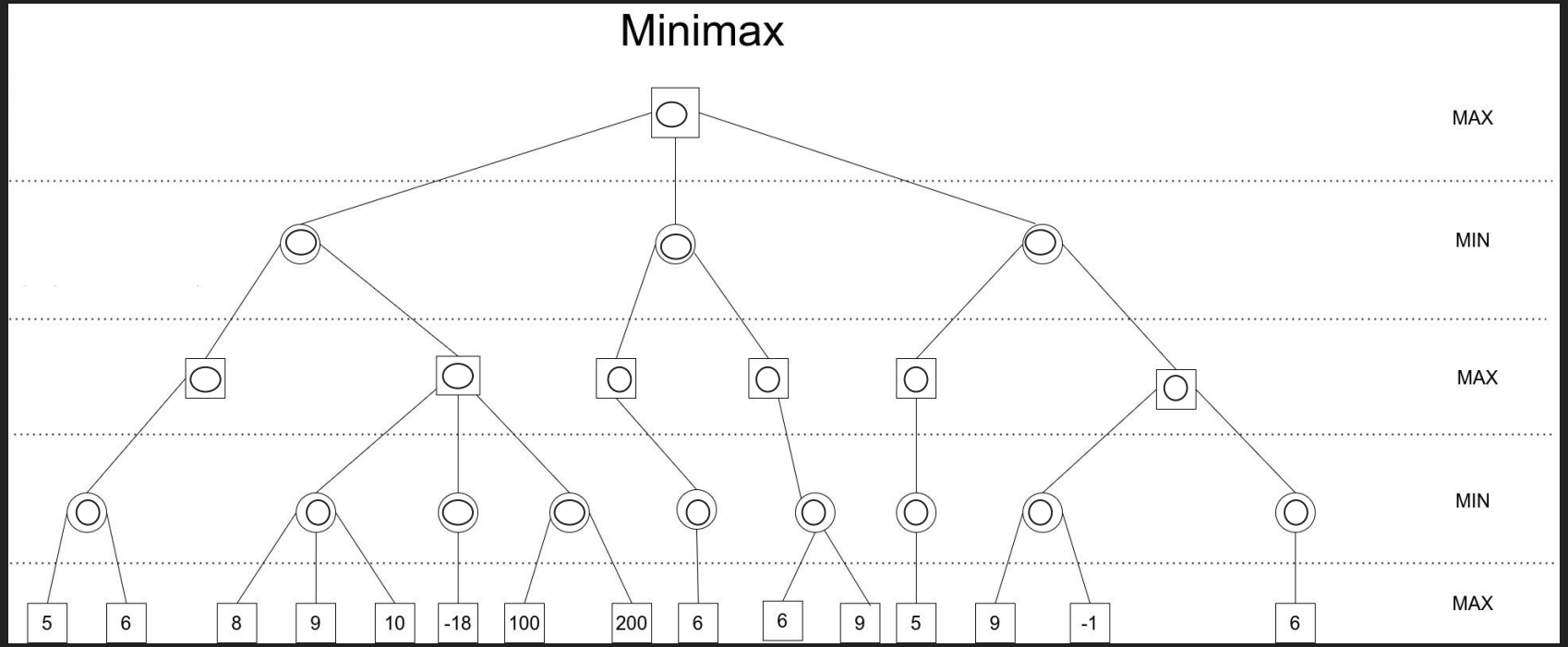
How can we evaluate and help AI make better decisions if we cannot see the end of the game?



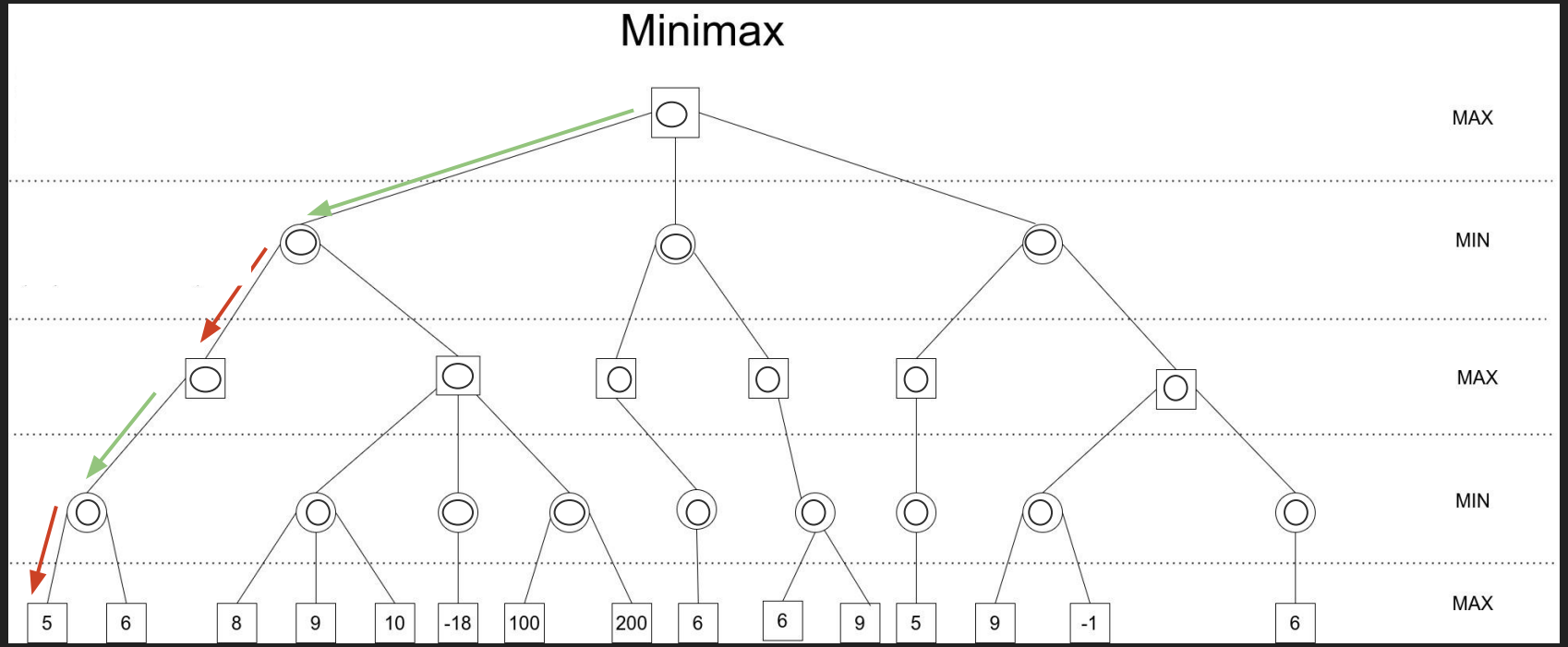




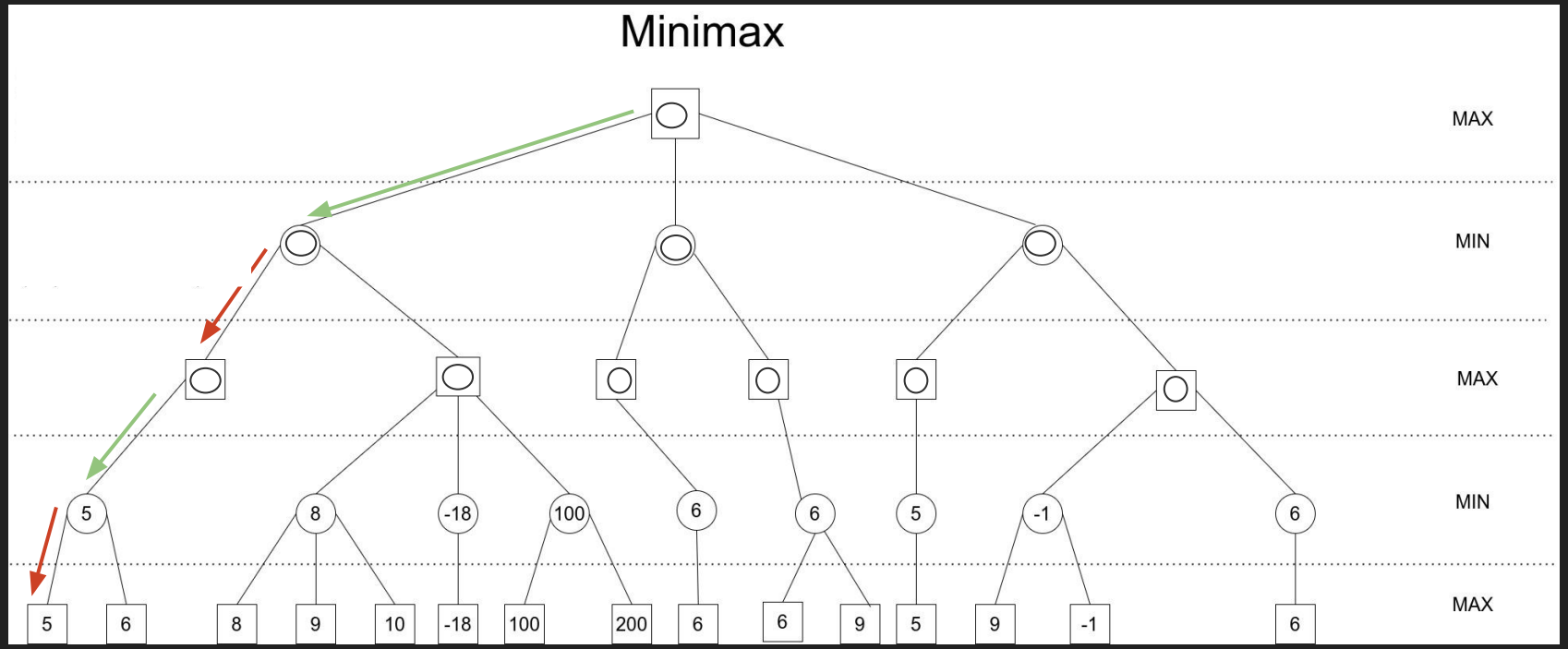
MiniMax Algorithm



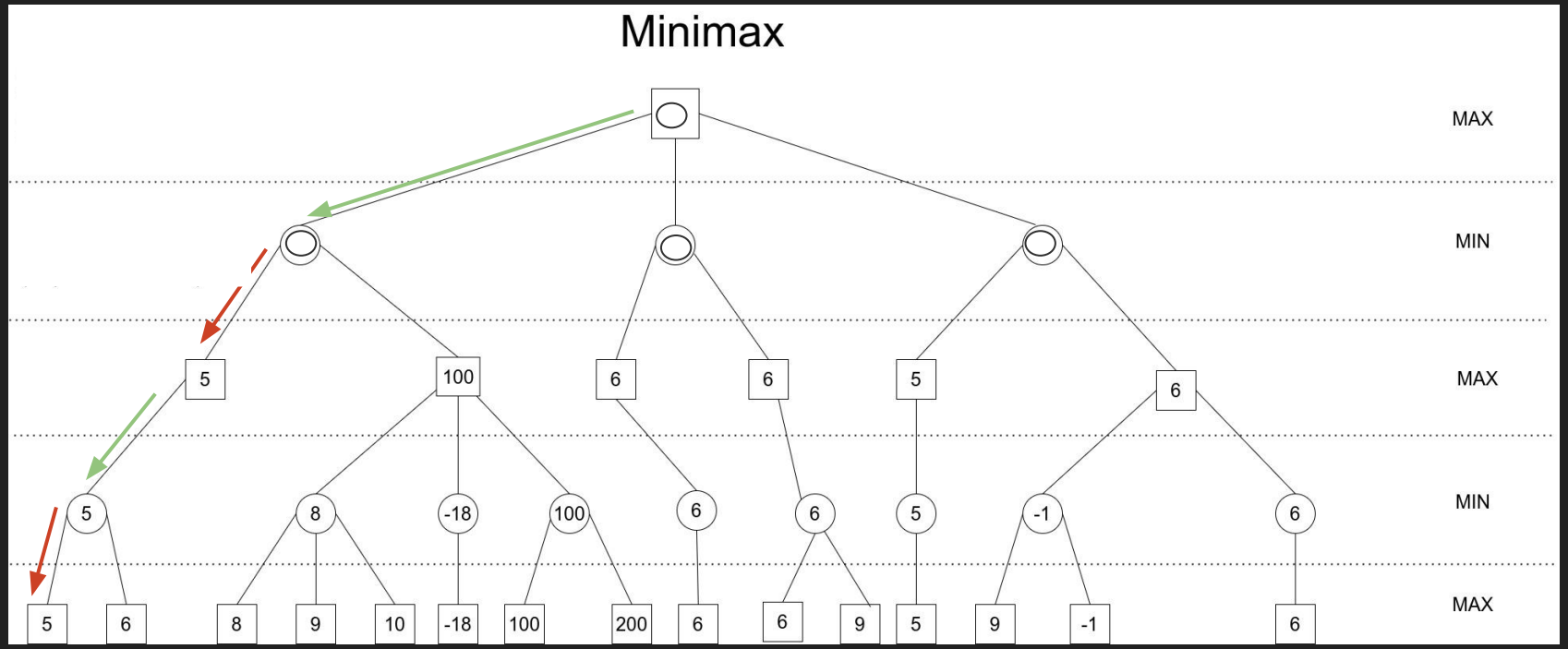
MiniMax Algorithm



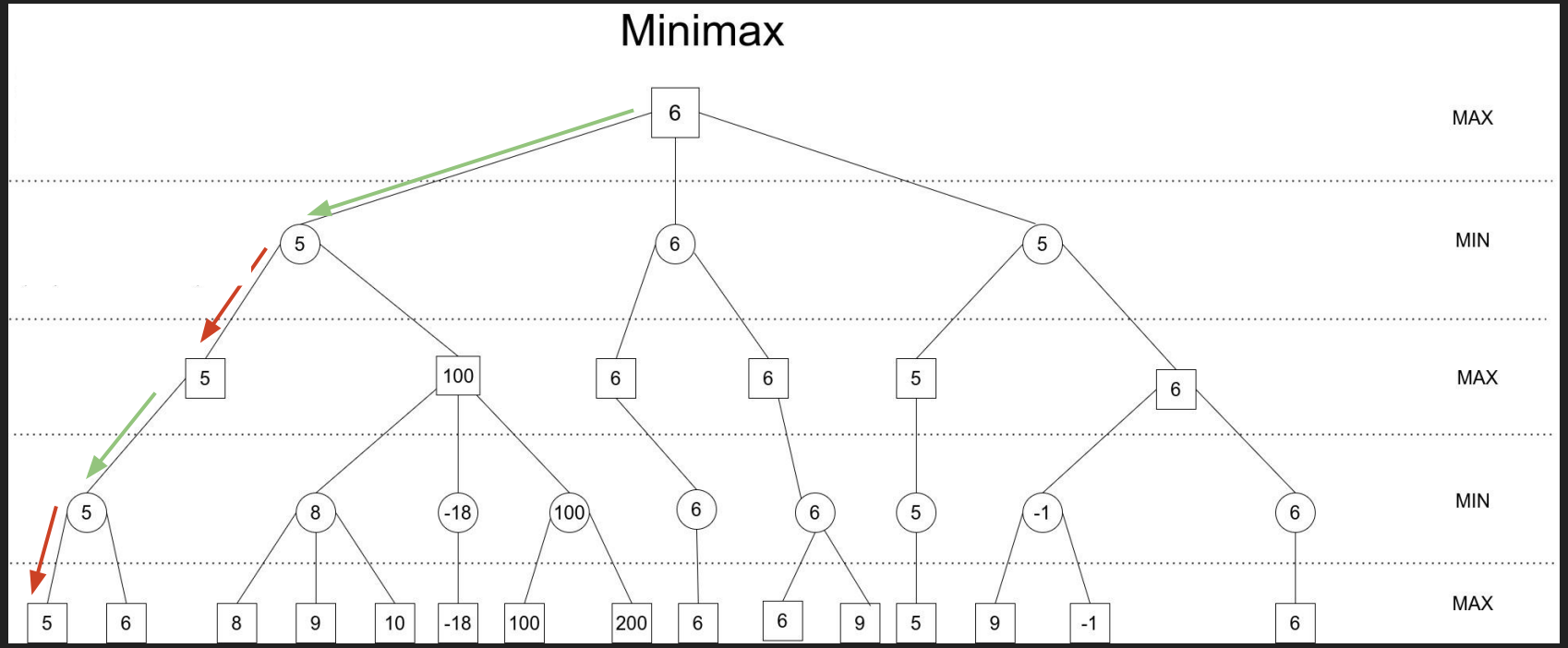
MiniMax Algorithm



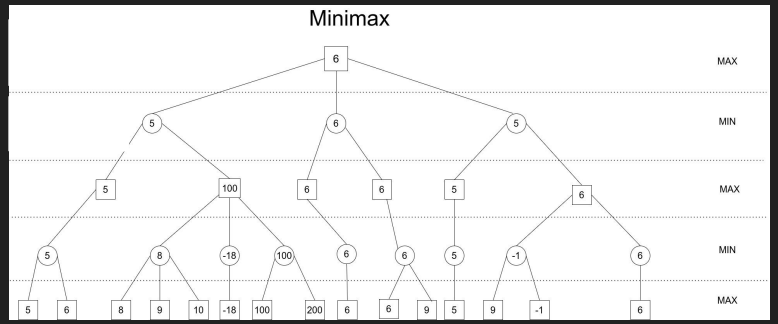
MiniMax Algorithm



MiniMax Algorithm



MiniMax Algorithm

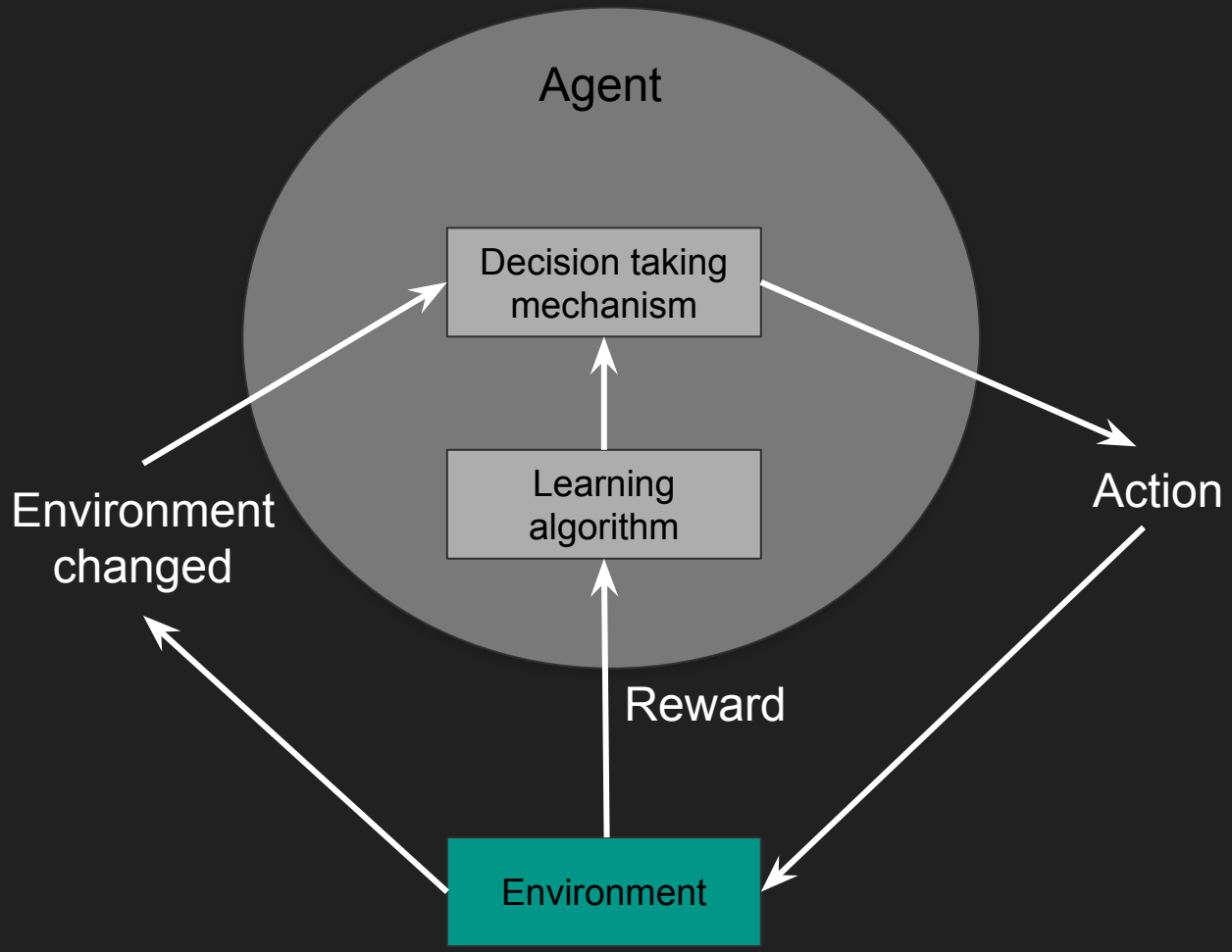



```
Minimax(node, depth, is_maximizing_player)
    IF depth > 3 OR node is terminal
        RETURN the heuristic value of node

    IF is_maximizing_player
        best_value = -∞
        FOR each child of node
            value = Minimax(child, depth + 1, FALSE)
            best_value = max(best_value, value)
        RETURN best_value

    ELSE
        best_value = ∞
        FOR each child of node
            value = Minimax(child, depth + 1, TRUE)
            best_value = min(best_value, value)
        RETURN best_value
```

What did we learn today?


Agent
Decision taking mechanism
Learning algorithm
Environment changed
Action
Reward
Environment

An exciting example of a very recent embodied AI

WAYVE

WAYVE

Alex Kendall (He/Him) · 2nd

CEO at Wayve

Education

University of Cambridge

Doctor of Philosophy (Ph.D.), Deep Learning, Computer Vision and Robotics

2014 - 2017

Co-Founder & CEO

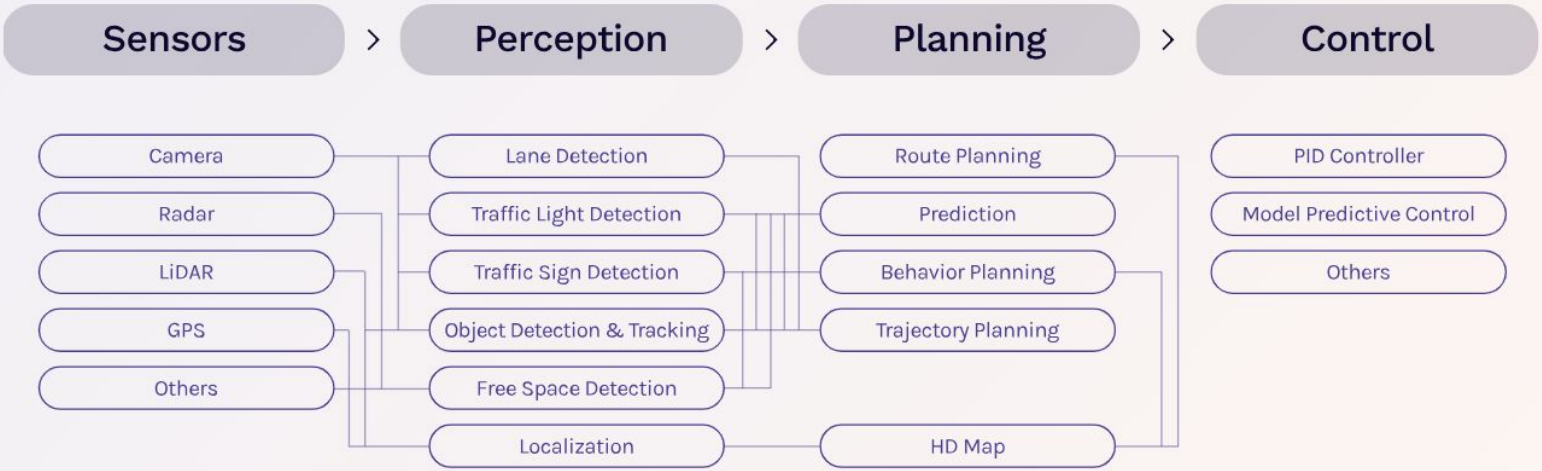
Wayve

Sep 2017 - Present · 7 yrs 7 mos

Wayve is building AV2.0—a next generation autonomous driving system that can quickly and safely adapt to new driving domains anywhere in the world. I firmly believe that large scale deployment of auton ...see more

WAYVE

AV1.0

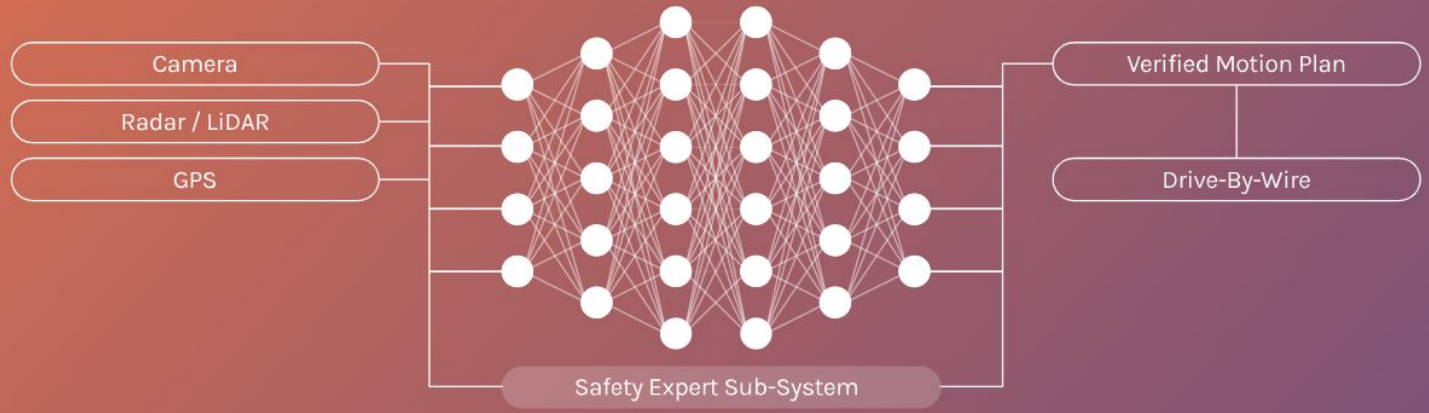

Sensors
Perception
Planning
Control
Camera
Radar
LiDAR
GPS
Others
Lane Detection
Traffic Light Detection
Traffic Sign Detection
Object Detection & Tracking
Free Space Detection
Localization
Route Planning
Prediction
Behavior Planning
Trajectory Planning
HD Map
PID Controller
Model Predictive Control
Others

AV1.0

Sensors › Perception › Planning › Control

AV2.0

Sensors › End-to-end AI › Control



GAIA-1:
A Generative World Model for Autonomous Driving

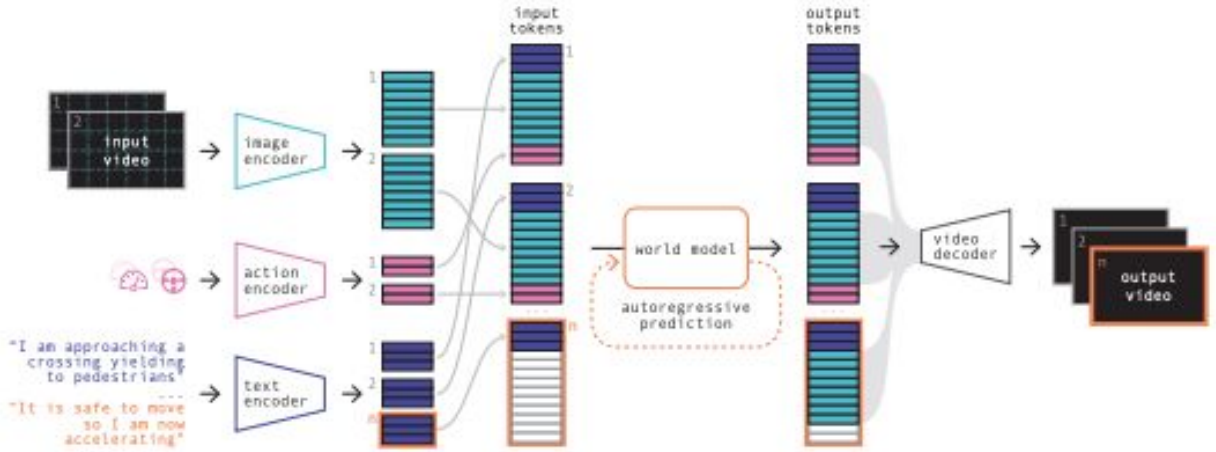


Figure 2: Architecture of GAIA-1. First, we encode information from all input modalities (video, text, action) into a common representation: images, text and actions are encoded as a sequence of tokens. The world model is an autoregressive transformer that predicts the next image token conditioned on past image, text, and action tokens. Finally, the video decoder maps the predicted image tokens back to the pixel space, at a higher temporal resolution.

Link: https://github.com/TeoReu/VO_RL